



Team 2: EV Battery Cooling and Cold Plates

Jonathan Buryj jburyj2@illinois.edu

Cadyn Grafton caydnjg2@illinois.edu

Iakov Gurevich iakovg2@illinois.edu

Chenxi Huang chenxih8@illinois.edu

Arjav Jain arjavj2@illinois.edu

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Abstract

Artificial intelligence (AI) and computational fluid dynamics (CFD) have become increasingly integrated over the past decade, driven by advances in deep learning and data-driven modeling that have matured into practical engineering tools. In thermal–fluid system design, a key limitation remains the high computational cost of high-fidelity CFD, which restricts both the breadth of multi-parameter design exploration and the speed of optimization.

This project applies an AI-assisted workflow to the design of an EV battery liquid cold plate, a widely used thermal management solution due to its high heat removal capability, efficient packaging beneath battery modules, and scalability in manufacturing. The cold plate design problem is inherently multi-objective: improvements in peak battery temperature and thermal uniformity typically come at the expense of increased pressure drop and pumping power. An optimal design must therefore balance thermal performance with hydraulic efficiency.

To address this trade-off, the class will conduct a coordinated design of experiments (DOE) study using a common baseline geometry and a standardized ANSYS Fluent conjugate heat transfer (CHT) setup. Nine groups will each perform 70 parametric CFD simulations using Latin hypercube sampling within defined subspaces, supplemented by shared corner and quality assurance cases. Key outputs, including maximum and average battery temperatures, temperature spread, pressure drop, pumping power, and thermal resistance, will be collected and combined into a unified dataset.

Using Python-based machine learning tools (e.g., NumPy, Pandas, scikit-learn, and gradient-boosted regression), surrogate models will be trained and validated to replicate CFD predictions with significantly reduced computational cost. These models will enable rapid constrained optimization and Pareto front analysis of thermal versus hydraulic performance, minimizing reliance on expensive CFD simulations while maintaining accuracy through selective validation.

1. Introduction

1.1 Literature Review

The rapid growth and transition towards electric vehicles (EVs) has placed lithium ion batteries at the forefront of automotive engineering and design. As the adoption of EVs has taken off through technological advancements in battery chemistry, progressive government policy, and increased consumer demand for fast-charging, powerful vehicles, the systems responsible for managing the battery's temperature have transformed from a second thought to a primary design constraint. Unlike historical batteries, lithium ion batteries are very sensitive to temperature, where optimal battery performance strongly depends on operating the device at a small temperature window. Straying away from this window brings forth a variety of risks, from battery degradation to dangerous thermal runaway. As charging rates and high-power vehicles continue to grow in importance, the thermal loads these batteries must handle will grow proportionately, making Battery Thermal Management System (BTMS) essential to keeping EVs a viable and safe option within the automotive market.

This literature review examines the current state of BTMS across four different areas. To begin, the safety, performance, and longevity of battery cooling are discussed to first establish why thermal management is critical to the system as a whole. Next, the historical development of lithium ion technology and the respective regulatory policy that boosted EV implementation are examined. Third, liquid cold plates, which are the dominant system in commercial EVs, are discussed in detail. Finally, alternative cooling approaches are compared to the liquid cold plate system to determine which cooling approach is most applicable to EV design.

1.1.1 Mission-Critical Battery Cooling

Battery cooling is a critical task in modern electric vehicle architectures. Today, as the industry moves toward extreme fast charging (XFC) and high energy density, the issue of battery cooling requires increasing attention. Tai et al. (2024) [1] point out that the optimal operating temperature range for lithium-ion batteries is very narrow, typically between 25°C and 40°C. Once this range is exceeded, the chemical stability of the battery is severely compromised, potentially leading to adverse consequences such as combustion or explosion. Furthermore, the future trends toward extreme fast charging (XFC) and high energy density make it imperative to address the issue of thermal runaway. Thermal runaway is a phenomenon that occurs when the internal heat generation rate exceeds the heat dissipation rate, triggering a chain reaction of exothermic reactions that cause the battery to reach extremely high temperatures in a very short time. Battery thermal runaway can lead to severe consequences. At the same time, a battery's cooling performance is also closely linked to its lifespan. Prolonged operation at high temperatures outside the ideal range can trigger thermally induced loss of active lithium, while also creating temperature gradients that cause uneven current distribution, thereby reducing the battery's service life. In extreme cases, this may even trigger thermal runaway, compromising safety. It is evident that battery cooling is a critical aspect of design and manufacturing, and the Battery Thermal Management System (BTMS) is increasingly becoming an indispensable "mission-critical" component in modern electric vehicle architectures.

From a safety perspective, the primary objective of battery cooling is to prevent thermal runaway. The essence of thermal runaway is that the rate of “heat generation” within the battery exceeds the rate of “heat dissipation” by the cooling system. Weng et al. (2019) [2] noted that differences in internal heat transfer rates and heating locations can lead to temperature inhomogeneity within the battery, with heating near the cathode causing the most intense combustion. This insight helps us effectively prevent the issue of localized hotspots. Ye et al. (2024) [3] experimentally demonstrated that immersion cooling can effectively suppress the propagation of thermal runaway between battery cells; increasing the immersion depth extends the time interval from safety valve rupture to thermal runaway initiation from 70 seconds to 312 seconds and significantly reduces peak temperatures. Therefore, when designing BTMS, it is essential to strictly control system temperature and optimize cooling technology to ensure that, under any operating conditions, the battery’s maximum temperature remains well below the critical threshold for thermal runaway.

From the perspective of fast-charging efficiency, effective battery cooling can enhance charging performance. In modern automotive design, fast-charging capabilities have gained popularity among users due to their significantly improved charging efficiency; however, this has also raised new challenges. Fast charging entails a substantial increase in current. Tai et al. (2024) [1] note that, according to Joule’s law, when current rises rapidly, the generated heat increases quadratically. If battery cooling is inadequate, the battery temperature can easily reach safety thresholds, causing the BMS to forcibly reduce the charging current and rendering the fast-charging mode ineffective. Therefore, to support reliable fast-charging performance, the BTMS must possess robust heat dissipation capabilities to limit the battery’s peak temperature.

From the perspective of battery lifespan, prolonged operation at temperatures above the ideal range can lead to battery degradation. Che et al. (2026) [4] point out that in parallel battery modules (PBMs), cell-to-cell variations and temperature gradients can cause significant branch current imbalance. This leads to pronounced temperature gradients within the battery, resulting in uneven current distribution, which further accelerates the thermal load on hot-spot cells and leads to costly warranty expenses. Therefore, to address this issue, another core design criterion for BTMS is to control temperature inhomogeneity, ensuring that all cells operate at consistent temperatures to achieve overall system reliability.

In summary, the cooling capacity of a battery system directly determines the safety, charging efficiency, and warranty lifespan of an electric vehicle. To ensure battery system performance, BTMS design must balance two key factors: limiting peak temperatures and minimizing temperature inhomogeneity. To meet this requirement, researchers are working to develop new optimization methods. As demonstrated by advanced technologies such as Direct Fluid Immersion Cooling (DFIC), studied by Suresh Patil et al. (2021) [5], this research confirms that direct contact cooling can reduce the maximum temperature by 46.8% compared to natural convection, highlighting a future direction for BTMS design.

1.1.2 Acceleration of BTMS

Throughout the past two centuries, rechargeable battery technology has drastically changed and evolved. More specifically, bulky lead-acid based cells have been converted to lithium-ion batteries that power modern electric vehicles. While initially, lead-acid based battery cells were adopted,

their low energy density and large weight made competitive driving to combustion motors infeasible [6]. The first adaptation of lithium ion technology occurred in the 1990s, where they were used in consumer electronics [7]. This ultimately set the framework for portable energy storage to be implemented into automotive design.

The transition from consumer electronics to automotive manufacturing was gradual to begin. Early EVs of the 1990s and 2000s all relied on lead-acid and NiMH packs, which were not capable of providing enough power to long-range passenger vehicles [6]. However, Lithium ion technology was first adopted into the Tesla Roadster in 2008, which was designed to hold thousands of cylindrical lithium ion cells and set the precedent that an EV vehicle was capable of matching the performance of higher-end sports cars [8]. From there, the automotive industry responded, where the Nissan Leaf (2010) became the first high-volume lithium ion passenger EV [8]. To compete with this, General Motors introduced the Chevrolet Volt, which was a plug-in hybrid [8]. Overall, these early developments established the foundation for battery management in passenger vehicles, and also birthed the framework for BTMS [8]. By 2018, EVs and electric busses now account for approximately 44% of the global lithium ion battery demand, which confirms the significance of BTMS in the industry [8].

The spread of EVs was not driven just from technological advancements alone; rather, supporting government policy played an equal role in public adoption. Emission regulations, purchasing incentives, zero-emission mandates, and infrastructure investments were the ultimate drivers to economically incentivize the transition to electric [9]. Within the United States, California's Zero Emission Vehicle program and the 2022 Inflation Reduction Act directly stimulated the demand for battery manufacturing within the US [10]. The European Union went even further, with a new mandate that all new passenger vehicles sold after 2035 must be zero emission [9]. This combined effort was a catalyst for the EV industry, and simultaneously raised additional engineering pressure to improve the performance, reduce the costs, and extend the lives of lithium ion batteries and BTMS.

As the consumer need for charging speed and vehicle performance increased, so did the thermal demands on these battery packs. The C-rate describes how quickly a battery can be charged or discharged relative to its total capacity [1]. 1C is equal to a full cycle in one hour, 2C in thirty minutes, and so forth [1]. As the need for higher C-rate charging became apparent, so did the associated costs with this new technology. The commercial roll out of DC fast charging brought new risks of thermal runaway and lithium ion cell degradation. Modern fast chargers that operate at 150-250kW can subject batteries to C-rates of 2C or higher, which can generate intense amounts of heat in a short amount of time [1]. Research shows that high C-rate, combined with elevated temperatures, can degrade a battery far faster than either of these stressors alone [1]. Additionally, high-performance driving platforms can compound onto this problem, as aggressive accelerating and braking conduct thermal stress over the battery's lifetime [7].

Overall, the evolution from lead-acid to high-density lithium ion EVs has been an engineering marvel and commercial landmark for global policy and scientific advancement. However, this progress has been met with the proportional increase in thermal demand and complexity. The growing need for faster charging and state-of-the-art vehicle performance pushes current EVs into high C-rate ratios, where heat generation and thermal engineering become a primary constraint in engineering design. Thus, making an efficient BTMS is paramount to ensuring the optimization of

the EV industry.

1.1.3 Dominance of Liquid Cold Plates

Liquids possess substantially higher thermal conductivity and heat capacity than air [11]; therefore, liquid cooling systems have proven to be capable of removing higher heat fluxes while improving temperature uniformity across battery modules compared to air-cooling technology [12]. For this reason, liquid-based thermal management systems have become the staple for high-power EV battery packs.

Cold plates provide an effective method for transferring heat from battery modules to the coolant. Typically placed directly beneath battery modules or cell arrays, the plates minimize thermal resistance between the heat source and coolant while enabling compact integration within battery pack structures. The microchannels within the cold plates improve convective heat transfer and temperature uniformity, since they increase the surface area to volume ratio of the thermo-conductive material (i.e. copper or silicon) and coolant used. Channel geometries such as serpentine-oscillating (Figure 1) disrupts the fluid boundary layer, placing the coolant near the surface to enhance heat transfer [13].

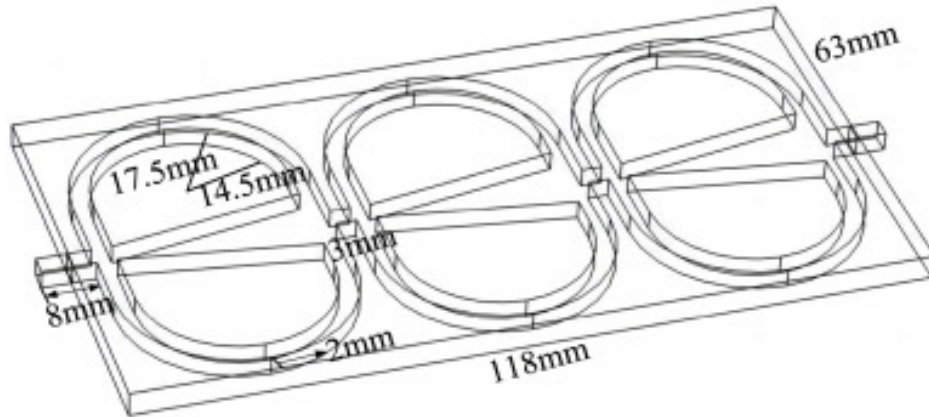


Figure 1: Schematic Structure of Serpentine-Oscillating Microchannel

Topology-optimized flow paths (Figure 2) have also been shown to significantly improve cooling effectiveness by controlling maximum temperature, reducing the average temperature, and suppressing the temperature gradient of the battery module [14].

Another important advantage of liquid cold plates is their compatibility with manufacturing processes. Cold plates are commonly fabricated from aluminum [14] using techniques such as extrusion, stamping, machining, brazing, or friction stir welding. These manufacturing methods are already widely used in automotive heat exchangers and power electronics cooling systems, allowing manufacturers to leverage existing supply chains and production expertise [11]. As a result, liquid cold plate designs can be produced reliably and cost-effectively at the large volumes required for EV battery pack production.

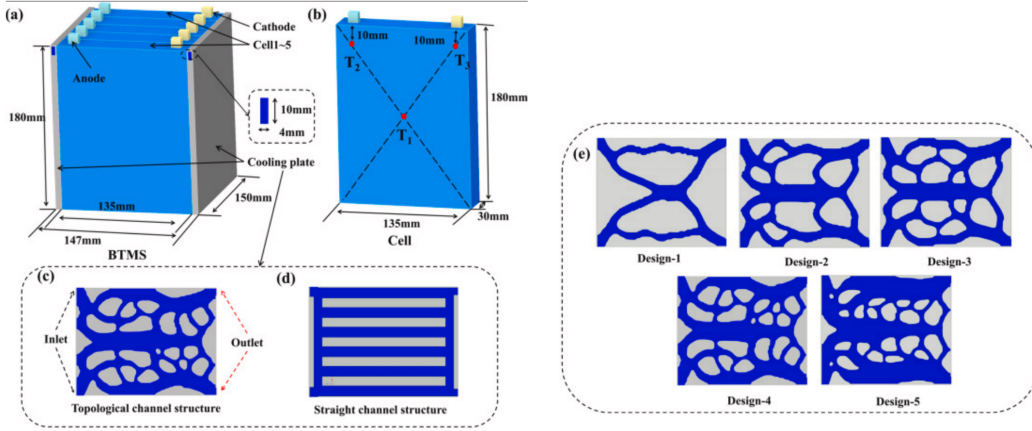


Figure 2: Schematic Structure of Topology-Optimized Microchannel

Liquid cold plates also integrate naturally with glycol–water coolant loops that are commonly used throughout vehicle thermal management systems. Many EV architectures employ centralized coolant loops that regulate the temperatures of batteries, electric motors, and power electronics. Cold plates connected to these loops allow engineers to control battery temperatures through adjustments in coolant flow rate, pump operation, and thermal routing strategies [11]. This controllability enables the battery pack to be maintained within an optimal temperature range during varying operating conditions, including rapid charging, high load operation, and cold-start scenarios. The ability to actively regulate battery temperatures is essential for maintaining electrochemical performance and extending battery lifetime.

1.1.4 Comparing Alternatives

According to Pesaran [15], an effective battery thermal management system (BTMS) must fulfill four primary functions: (i) cooling to dissipate excess heat generated during operation, (ii) heating to maintain adequate performance under low-temperature conditions, (iii) insulation to minimize rapid temperature fluctuations, and (iv) ventilation to safely remove any hazardous gases emitted from the battery. Furthermore, for practical implementation in electric vehicles, BTMS should exhibit additional desirable attributes, including compact design, low weight, cost-effectiveness, high reliability, ease of maintenance, minimal parasitic energy consumption, and efficient integration within the vehicle architecture [16, 17].

Several review studies on battery thermal management systems (BTMS) have provided diverse perspectives on classification and performance. Rao and Wang [18] categorized BTMS based on heat transfer media: air, liquid, and phase change materials (PCM) with a focus on PCM behavior and associated trade-offs, extending the framework of Cosley and Garcia [19]. Wang et al. [20] further expanded this by examining both intrinsic battery thermal characteristics and external cooling strategies, classifying systems into air-, liquid-, PCM-, heat pipe-based, and hybrid configurations. Xia et al. [21] adopted a multiscale approach, analyzing thermal behavior at the cell level and evaluating BTMS performance at the module level, including comparisons of air and liquid cooling under various cell arrangements, with liquid systems further divided into tube- and plate-based designs using mini/micro-channels.

1.1.5 Battery Cooling

A battery thermal management system (BTMS) can be classified as passive or active based on the mode of heat removal, with passive systems relying on ambient conditions and active systems employing externally driven cooling media. Additionally, BTMS can be categorized as external or internal depending on the location of heat extraction [22]. Most existing approaches are external, where heat is removed from the battery surface. However, due to thermal resistance between the cell core and surface, peak temperatures typically occur internally, and increasing surface convection can further intensify temperature gradients [23]. To overcome this limitation, internal cooling strategies have been proposed, enabling direct heat removal from within the cell and improving temperature uniformity [24]. For example, Mohammadian et al. [25] embedded microchannels within the electrode region, using the flowing electrolyte as a coolant. Given that heat generation is highest near the electrodes [26], this method significantly enhances thermal uniformity, reducing temperature variation by more than fivefold at a pumping power of 0.024 W. Despite these advantages, internal cooling remains limited in practice due to the structural constraints of sealed battery designs.

1.1.6 Liquid Cooling

Liquid cooling systems can be categorized into indirect and direct (immersion) cooling, depending on whether the coolant is in physical contact with the battery cells, as shown in Fig. 3.

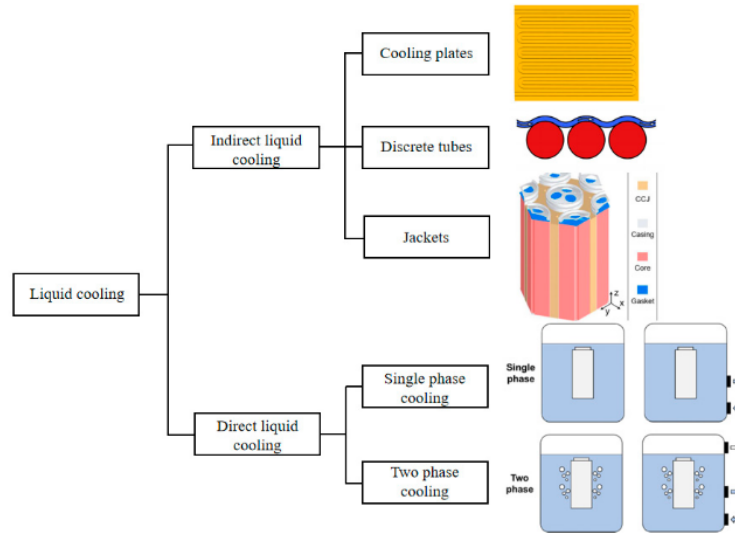


Figure 3: Classification of liquid cooling methods and corresponding examples [27]

Direct liquid cooling (immersion cooling) involves direct contact between the battery and a dielectric coolant, significantly enhancing thermal performance. The most notable advantage of this method is the elimination of interfacial thermal resistance, which leads to substantially higher heat transfer coefficients and improved temperature uniformity. Studies [28, 29, 30] have shown that immersion cooling can achieve lower maximum temperatures and reduced temperature gradients compared to indirect systems, particularly under high discharge rates. Additionally, direct cooling enables more compact and lightweight system designs, as it removes the need for bulky cold plates

and complex flow channels. It also demonstrates strong potential in thermal runaway mitigation, especially when two-phase coolants are used, due to the high latent heat absorption during boiling. However, these advantages come with significant drawbacks. The requirement for dielectric fluids (e.g., silicone oil, fluorocarbons) increases material costs and introduces compatibility challenges with battery components. Moreover, such fluids typically exhibit higher viscosity, leading to increased pumping power and energy consumption compared to indirect systems [31, 32]. Additional concerns include coolant leakage risks, fluid management complexity, long-term reliability, and limited industrial adoption. In the case of two-phase systems, issues such as fluid volatility, cost, and system sealing complexity further restrict practical implementation.

Indirect liquid cooling is the most widely adopted approach in commercial electric vehicles due to its balance between performance, safety, and manufacturability. In this configuration, the coolant flows through cold plates, tubes, or jackets that are thermally coupled to the battery surface without direct contact [33, 34]. One of its primary advantages is electrical safety, as the coolant is isolated from the battery, eliminating risks of short circuits [35]. Additionally, mature coolants such as water or water–glycol mixtures offer low viscosity, high specific heat capacity, and relatively low pumping power requirements, resulting in lower parasitic energy consumption and proven reliability in automotive applications. However, indirect cooling suffers from thermal contact resistance due to the presence of interfaces (e.g., air gaps, thermal pastes), which reduces heat transfer efficiency [36, 37]. Consequently, temperature gradients within the cell remain significant, particularly at high C-rates. Furthermore, achieving effective cooling often requires complex channel designs and precise manufacturing, increasing system weight, cost, and maintenance difficulty. While optimization of flow paths and coolant properties can improve performance, the fundamental limitation lies in the indirect nature of heat extraction.

1.1.7 Composite Cooling

Phase Change Material (PCM) based BTMS have gained significant attention due to their ability to utilize latent heat for thermal regulation. Materials such as paraffin absorb and store heat during phase transition, acting as a thermal buffer that maintains battery temperatures within the optimal range while ensuring high temperature uniformity. Their passive operation eliminates the need for additional power input, improving overall system efficiency. However, the practical application of pure PCM is constrained by its inherently low thermal conductivity and potential leakage in the liquid state [38]. To address these limitations, recent research [21] has focused on composite PCMs, incorporating high-conductivity additives such as expanded graphite, carbon fibers, or graphene, which substantially enhance thermal performance. Furthermore, hybrid systems combining PCM with active cooling methods enable effective heat removal and material re-solidification under high-load or high-temperature conditions. In parallel, heat pipe–based cooling has demonstrated notable advancements, with studies [39] reporting reductions in battery surface temperature of up to 32 °C and significantly lower thermal resistance compared to natural and forced convection systems. Despite their high effective thermal conductivity, heat pipes are often limited by the restricted contact area with battery cells, necessitating integration with conductive structures such as cooling plates, particularly for prismatic and pouch cells.

To overcome the individual limitations of PCM and heat pipes, recent developments increasingly focus on hybrid configurations that combine both technologies. The primary strength of com-

posite systems lies in their ability to simultaneously address multiple thermal challenges, such as peak temperature reduction, temperature uniformity, and energy efficiency [40, 41, 42]. Moreover, composite systems can be tailored to specific operational requirements, offering greater design flexibility than single-method approaches. Despite these advantages, composite cooling systems introduce increased design complexity, integration challenges, and higher initial costs. The interaction between different cooling mechanisms must be carefully optimized. Additionally, issues such as added weight, volume constraints, and control system complexity can offset some performance gains. In certain configurations, the combined system may also suffer from diminishing returns, where the added complexity does not proportionally enhance cooling performance.

1.1.8 Literature Review Conclusion

Liquid cold plates remain the industry baseline because they offer the best balance between high heat removal capability, manufacturability, cost, and system-level integration, despite not being the most thermally optimal solution. Ongoing research is therefore driven by tradeoffs such as interfacial thermal resistance, pressure drop, and temperature non-uniformity, which limit performance under extreme operating conditions.

1.2 CFD and Modeling

The design process follows a structured workflow to develop and optimize an EV battery liquid cold plate using both CFD and data-driven methods. It begins with geometry creation and mesh independence analysis, where the physical configuration of the cold plate is defined and discretized. A mesh sensitivity study ensures that simulation results are not influenced by grid resolution, establishing numerical accuracy with minimal computational cost. Next, parametric conjugate heat transfer (CHT) simulations are performed. These simulations capture the coupled heat transfer between the fluid and solid domains, accounting for conduction within the plate and convection in the coolant. Key geometric and operating parameters are varied to understand their influence on thermal and hydraulic performance. The workflow then proceeds to the design of experiments (DOE) stage, where a systematic sampling strategy (e.g., Latin hypercube sampling) is used to explore the design space efficiently. This step generates a diverse dataset that captures the relationship between input parameters and performance metrics. Using this dataset, surrogate models are developed. These data-driven models approximate CFD predictions with significantly reduced computational time, enabling rapid evaluation of new design points without running full simulations. Finally, the models undergo training, testing, and validation to ensure predictive accuracy and generalizability. Once validated, the surrogate models can be used for optimization and Pareto analysis, balancing competing objectives such as thermal performance and pressure drop.

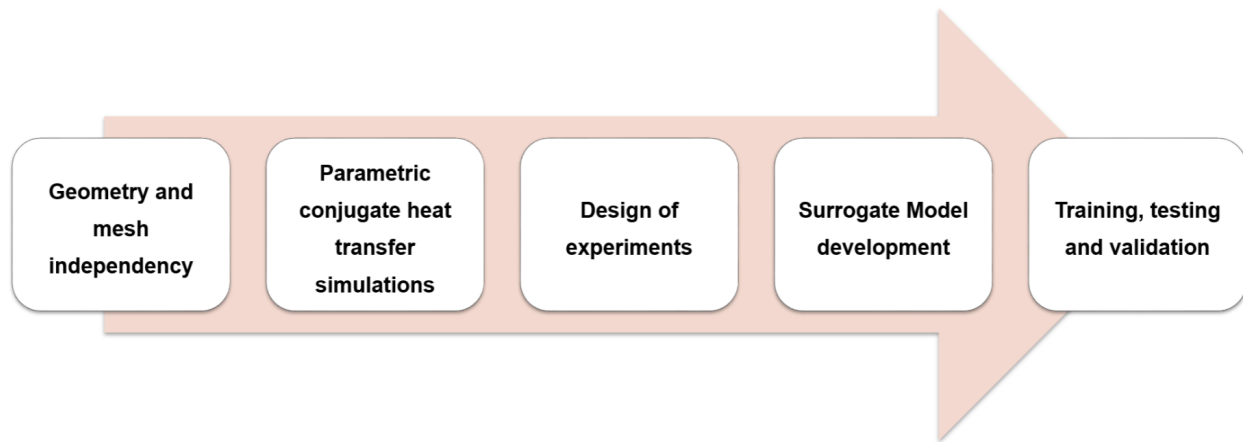


Figure 4: Design Process

2. Simulation Methodology

To evaluate the thermal and hydraulic behavior of the liquid-cooled EV battery cold plate designs, a comprehensive CFD-based simulation methodology was developed. The following subsections describe the modeling assumptions, numerical setup, and post-processing techniques used to investigate how channel geometry and coolant flow conditions influence cooling performance, pressure losses, and overall system efficiency.

2.1 Geometry & CFD Setup

Our team modeled steady state heat transfer in a liquid-cooled aluminum 6061 cold plate for an EV battery. The plate is fixed at a length of 100 mm, width of 40 mm, and thickness of 4 mm, with straight coolant channels running along the length. Only four inputs change: channel height, width, rib thickness, and total mass flow rate. The CFD setup uses steady, pressure-based Fluent with SIMPLE coupling, a laminar flow assumption, coolant consisting of both water and ethylene glycol. We have boundary conditions including a mass-flow inlet set at 25°C and an outlet pressure of 0. The top surface of the plate is set at a constant temperature of 40°C, while all other surfaces are assumed to be adiabatic. Our outputs include coolant temperature, pressure drop, and max velocity, and from these we are able to calculate pumping power, total heat removed, and average thermal resistance.

Using the provided channel heights and widths for each geometry, we solved for the hydraulic diameter of each geometry's channels. We then use this value to solve for the Reynolds number. The density and viscosity values are constant at 1072 kg/m^3 and $0.00353 \text{ kg/(m * s)}$ respectively, and we added a maximum velocity plot as an output at each iteration through Fluent as the model converged. After calculating the Reynolds Number for each of our 50 geometries tested, our range is between 34 and 475. This is well below 2300, the transition phase between laminar and turbulent flow. So every Team case is safely laminar, which supports the assignment's laminar model assumption.

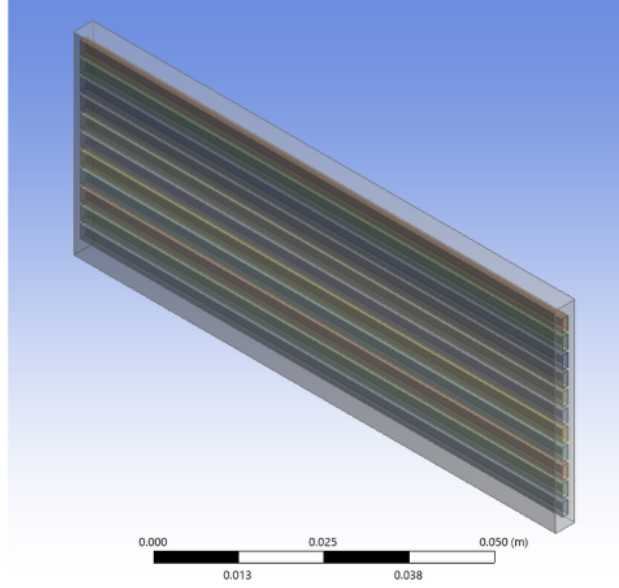


Figure 5: CFD setup

The Outlet Temperature vs Channel Height plot, in Figure 6, shows a weak relationship, with temperatures broadly scattered across all height values. The Channel Width plot on the right also exhibits significant scatter or variability of results. Wider channels seem to correspond to lower temperatures, but the trend is inconsistent. Because both height and width simultaneously affect several parameters, like the hydraulic diameter, flow area, and velocity distribution, combining them into a single nondimensional parameter provides a more physically meaningful description of the relationship. The Aspect Ratio plot on the left shows lower outlet temperatures tend to occur at moderate-to-higher aspect ratios, while higher temperatures are more concentrated at lower aspect ratios. Although some scatter remains, the data appears more organized compared to the individual geometry variables, and suggests that thermal performance depends more strongly on the relative shape of the channel rather than either dimension independently.

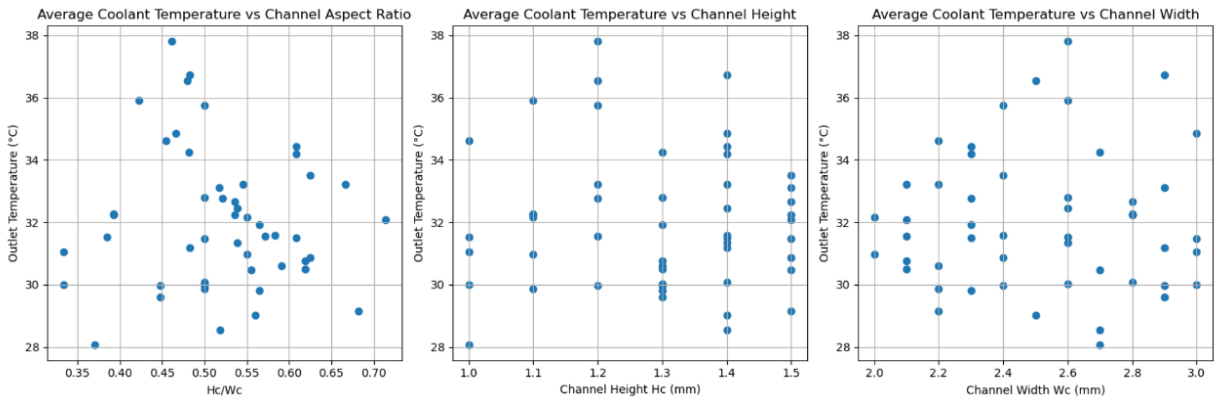


Figure 6: Channel aspect ratio (temperature)

The pressure-drop results show a stronger dependence on channel geometry than the temperature

results. Channel height exhibited the clearest influence on pressure loss, with larger H_c values producing substantially lower pressure drops. Wider channels also reduced pressure drop, with greater visual variability. While the aspect ratio captured the overall behavior, the individual channel dimensions—particularly channel height—were more predictive of hydraulic performance. This behavior is consistent with classical internal-flow theory, where smaller hydraulic diameters produce larger viscous losses and therefore higher pressure drops.

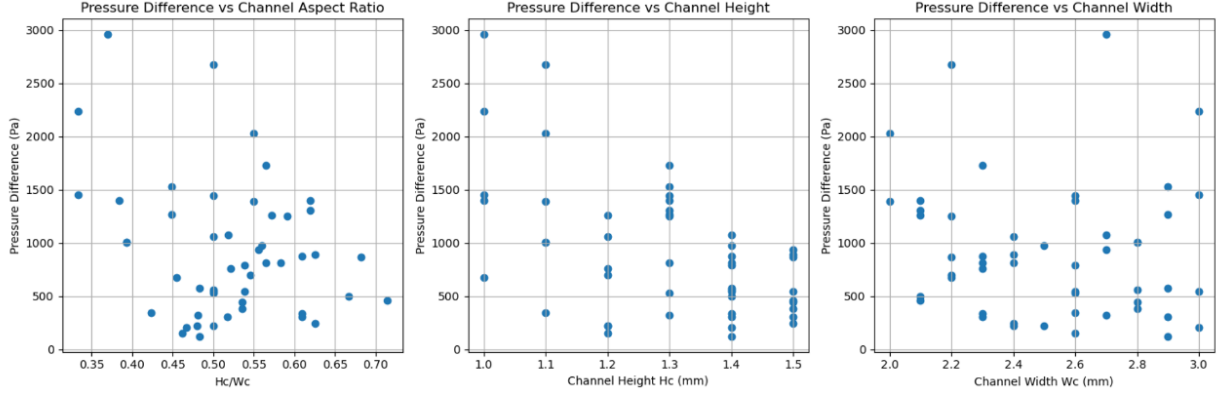


Figure 7: Channel aspect ratio (pressure)

Pumping power exhibited a strong dependence on channel geometry, particularly channel height. Increasing H_c generally reduced pumping power significantly due to reduced viscous resistance and lower pressure losses within the coolant channels. Wider channels also tended to lower pumping power. While channel aspect ratio captured some combined geometric effects, channel height alone appeared to be the strongest predictor of hydraulic cost.

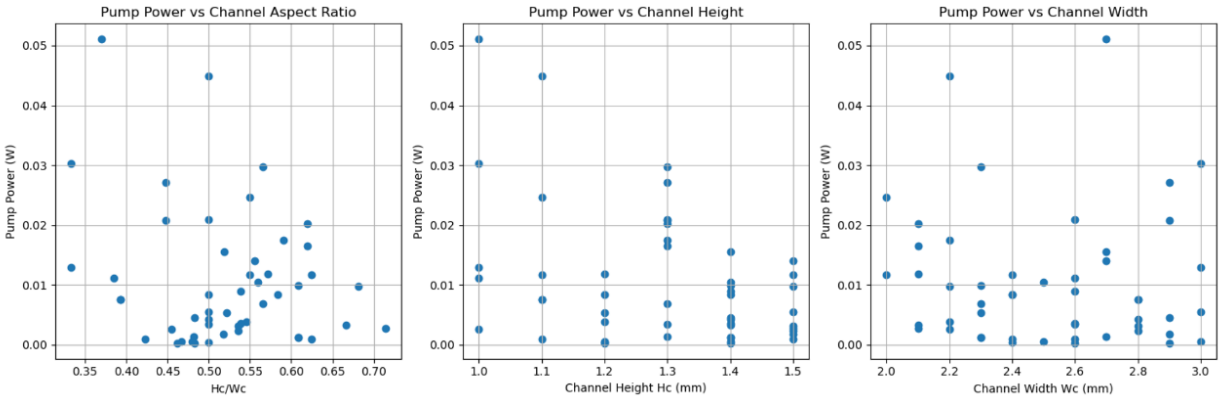


Figure 8: Channel aspect ratio (pump power)

The heat-transfer-rate results exhibited more complex behavior than the hydraulic metrics. Moderate-to-high aspect ratios generally produced improved heat-transfer performance, while channel width alone showed relatively weak predictive capability. Many of the highest Q values occurred for channel heights between 1.3–1.5 mm, suggesting that reduced hydraulic resistance and improved

coolant transport benefited overall heat removal. Compared with the individual geometry variables, the aspect ratio produced somewhat more organized trends.

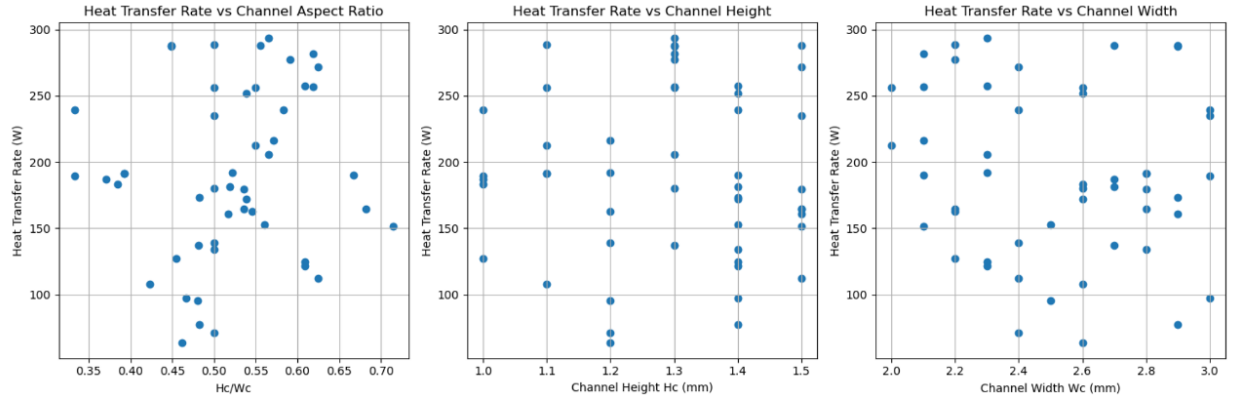


Figure 9: Channel aspect ratio (heat transfer)

Thermal resistance exhibited the clearest dependence on channel aspect ratio. Increasing the ratio generally reduced resistance, indicating improved cooling effectiveness at moderate-to-high aspect ratios. Channel height showed some influence, and channel width by itself once again produced relatively weak trends. Combined interaction governs thermal performance more effectively than either parameter independently.

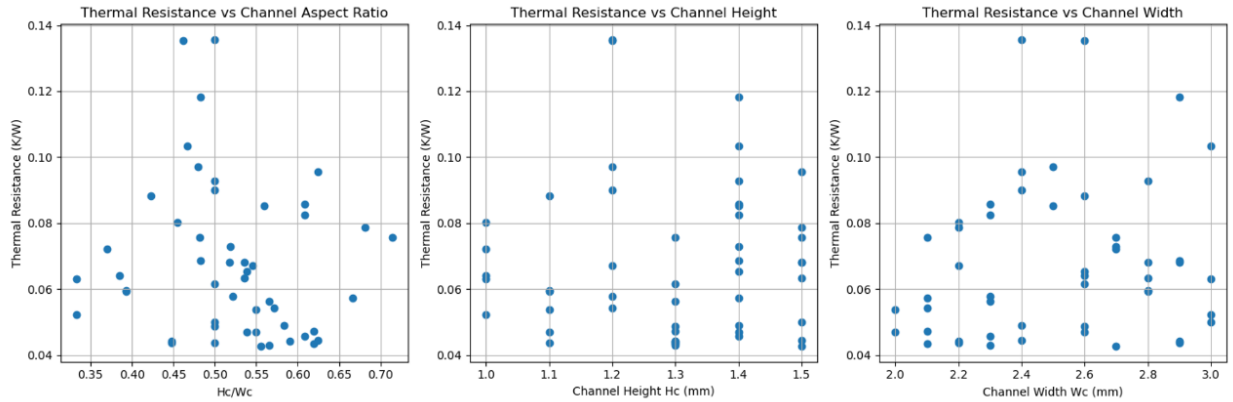


Figure 10: Channel aspect ratio (thermal resistance)

Temperature and velocity contour plots (Figures 11-14) for two of our simulations illustrate the influence of mass flow rate on both thermal and hydraulic performance. The low-flow-rate case on the left exhibited a larger temperature rise and stronger thermal gradients along the channel length, while the high-flow-rate case maintained more uniform coolant temperatures and improved overall cooling performance. The velocity contours confirmed substantially larger coolant velocities for the high mass flow case, which improves heat transfer but also increases viscous losses and pressure drop.

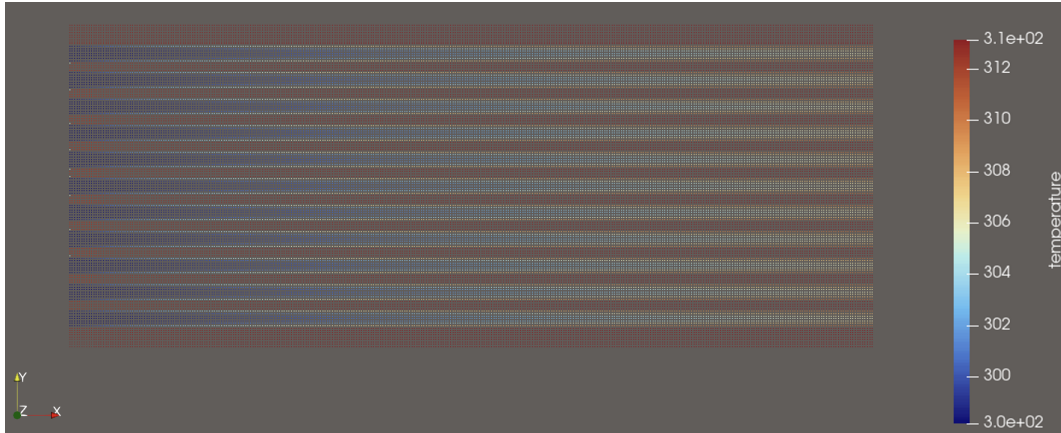


Figure 11: Temperature field (low mass flowrate)

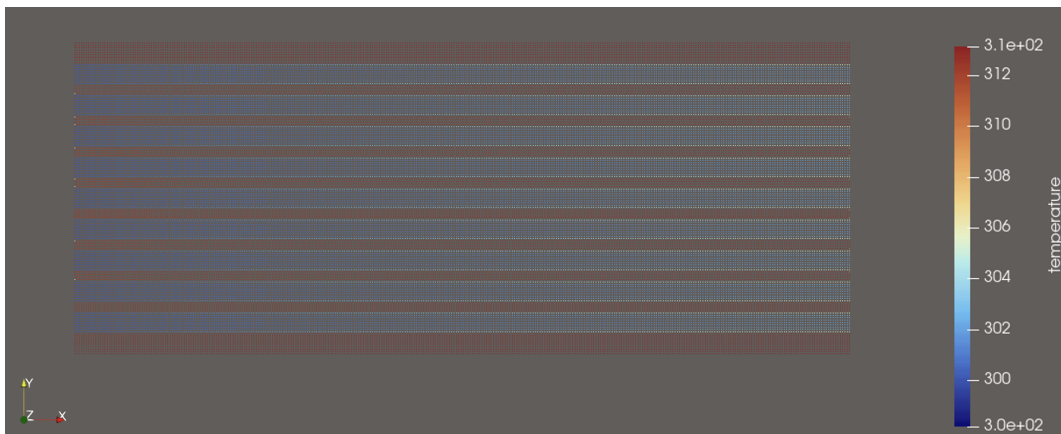


Figure 12: Temperature field (high mass flowrate)

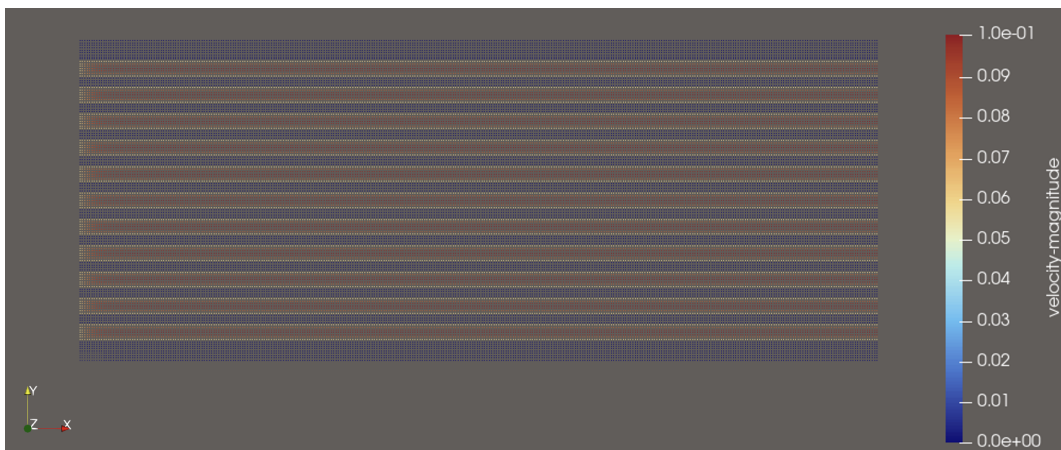


Figure 13: Velocity field (low mass flowrate)

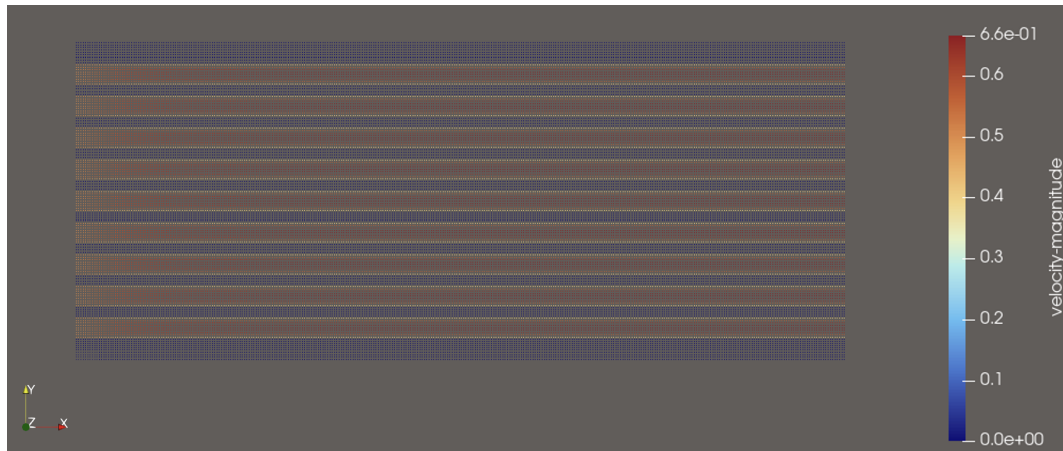


Figure 14: Velocity field (high mass flowrate)

These next set of contours (Figures 15-18) demonstrate the influence of channel width on coolant behavior. The narrower-channel case of 2.0 mm produced larger coolant velocities because the same mass flow was forced through a smaller flow area. The increased velocity enhanced local convective heat transfer and produced stronger thermal gradients along the channel length. The wider-channel case at 3.0 mm distributed the flow more evenly and reduced average velocity, leading to slightly more uniform temperature fields and lower expected hydraulic losses. Narrower channels improve convection but increase pressure drop and pumping power.

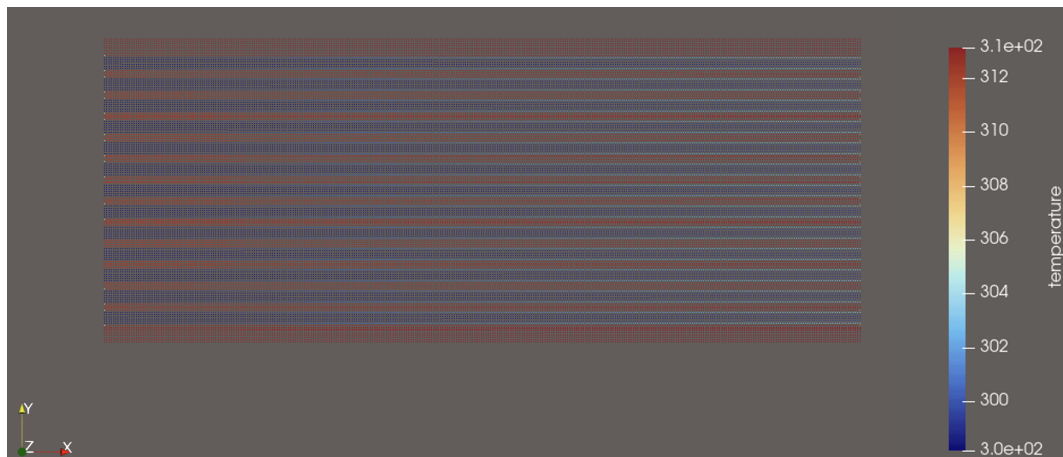


Figure 15: Temp field (low channel sizing)

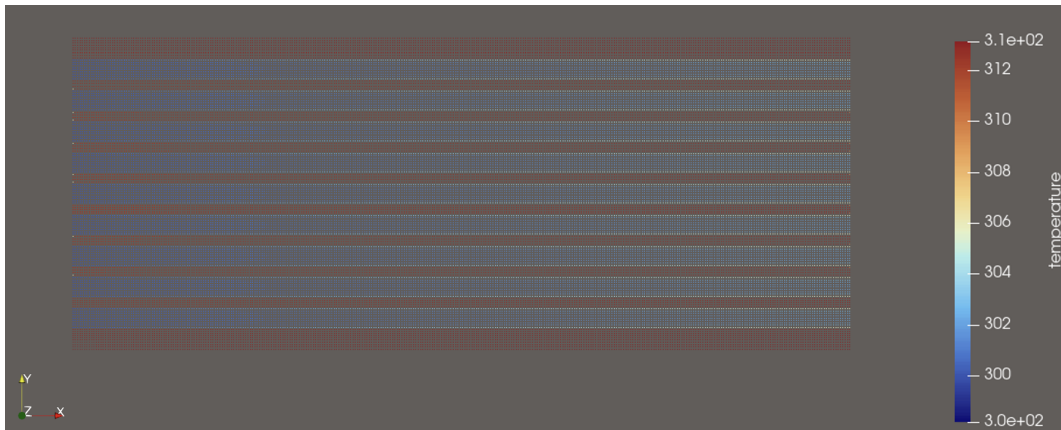


Figure 16: Temp field (high channel sizing)

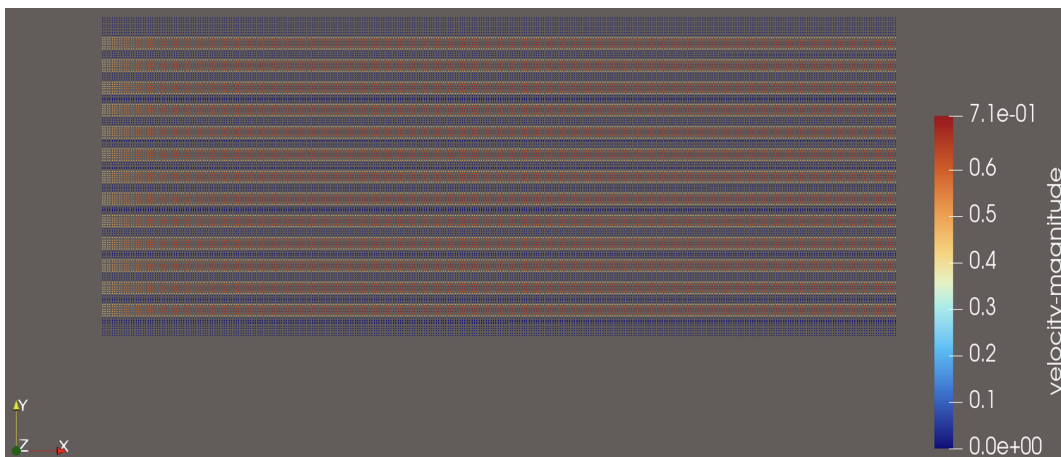


Figure 17: Velocity field (low channel sizing)

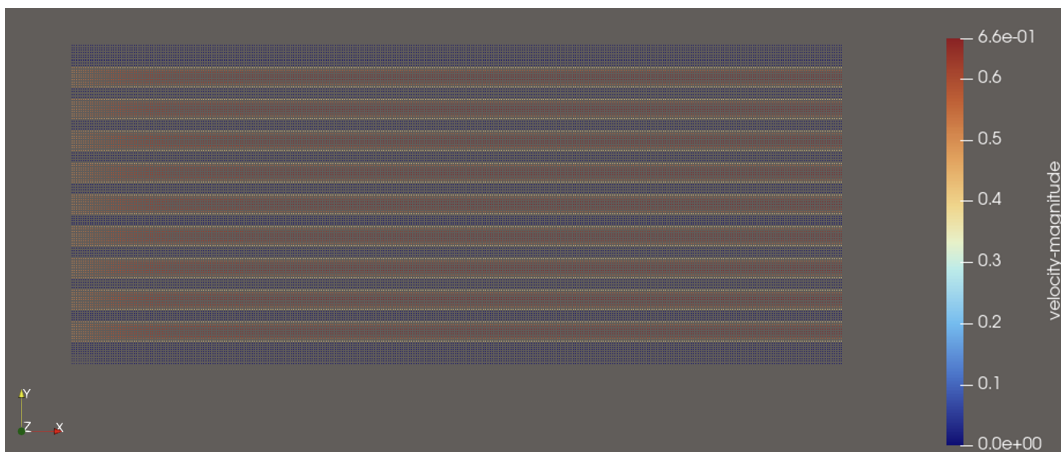


Figure 18: Velocity field (high channel sizing)

A python script was created with normalization equations for both pump power and thermal resistance, and the results are shown in the bottom right. A value of 0 corresponds to the best-performing case for that metric, and value of 1 corresponds to the worst-performing case. After adding the normalized values for each of our 50 simulations and comparing scores (Figure 19), it can be observed that LHS69 is the most optimal case, minimizing thermal resistance without excessive pumping power.

runid	P_pump_W	R_th_avg_KW	Rth_norm	Ppump_norm	Score
LHS_069	0.005545	0.050053	0.080195	0.104837	0.185032
LHS_049	0.008873	0.047003	0.047422	0.170191	0.217613
LHS_074	0.003268	0.057320	0.158273	0.060103	0.218376
LHS_016	0.009837	0.045700	0.033423	0.189125	0.222548
LHS_084	0.008362	0.048948	0.068318	0.160144	0.228462
LHS_045	0.011621	0.044439	0.019880	0.224156	0.244036
LHS_099	0.005301	0.057825	0.163704	0.100043	0.263747
LHS_085	0.003432	0.061610	0.204375	0.063328	0.267704
LHS_002	0.013977	0.042589	0.000000	0.270421	0.270421
LHS_015	0.006838	0.056131	0.145504	0.130224	0.275728

Figure 19: Optimal Case (LHS_69)

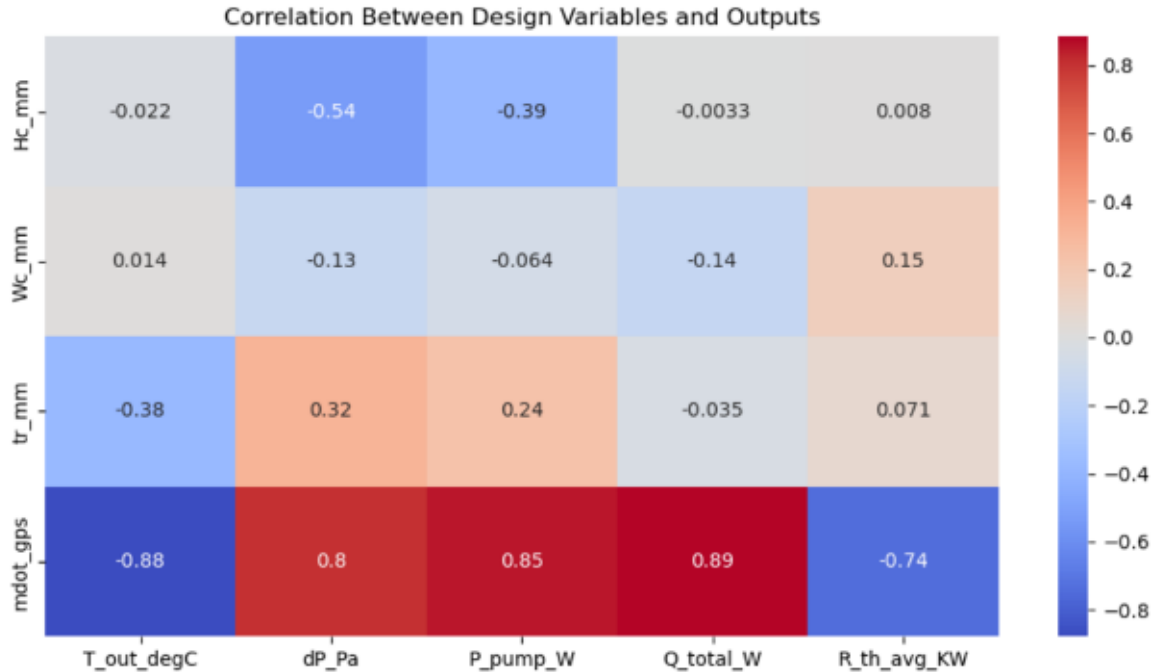


Figure 20: Correlation Table

A correlation plot (Figure 20) was also developed between variable inputs and desired output val-

ues. From this we determined that the coolant mass flow rate is by far the most dominant design variable in our design region. It exhibits the strongest correlations with almost every output metric, indicating that flow rate controls both thermal and hydraulic behavior. Among the geometry parameters, channel height has the strongest influence on hydraulic behavior, followed by rib thickness, and then channel width with the weakest influence overall.

2.2 Mesh Independence Study

CFD results are only as trustworthy as the mesh on which they are computed on. A coarser mesh introduces discretization errors, as thermal and pressure gradients are smeared across larger cells, which results in less accurate physical predictions. On the other hand, decreasing the mesh size to a finer resolution introduces a computational and memory tradeoff, where very fine meshes may become infeasible to compute due to hardware constraints. Thus, identifying the coarsest mesh size that provides sufficiently accurate results is essential to determine before conducting a full simulation campaign.

To determine the optimal mesh size, a mesh independence study was performed by running multiple simulations on the same geometry at progressively finer element sizes, ranging from 0.8mm to 0.2mm. The inlet pressure and outlet pressure were monitored as the key outputs across different mesh refinements. The results are summarized in Figure 21.

Mesh Independence Study					
Mesh Size (mm)	T_{out} (K)	P_{inlet} (Pa)	Converges?	ΔP (Pa)	% Change in ΔP
0.8	301.90	406.69	Yes	—	—
0.6	303.37	507.61	Yes	100.92	24.815
0.4	304.55	576.99	Yes	69.38	13.668
0.3	305.03	614.18	Yes	37.19	6.446
0.25	305.16	635.92	Yes	21.74	3.419
0.2	305.24	656.08	Yes	20.16	3.073

Figure 21: Mesh Independence Table

As shown in Figure 21, none of the tested mesh sizes produced diverging solutions. All cases reached the required residual criteria defined from the problem statement, and yielded stabled outlet temperature and inlet pressure values. This is physically consistent with the laminar Reynolds number regime and the relatively simple channel geometry of the cold plate. However, at coarser mesh sized, large jumps in the inlet pressure were observed between consecutive refinements, indicating that the solution remained sensitive to mesh resolution in that range. As the mesh was refined from 0.6mm to 0.3mm, the percent change in inlet pressure decreased from approximately 24.8% to 6.4%, which suggests that the solution was approaching mesh independence.

To investigate whether further refinement below 0.3 mm was warranted, mesh sizes of 0.25 mm and 0.2 mm were tested using a professional ANSYS license, where both produced percent changes in inlet pressure below 3.5%. However, when attempting these same mesh sizes on the EWS FastX computing environment, the meshing process would not reliably finalize, introducing a computational barrier. As a result, these finer mesh sizes were not viable for the scope of this project.

Based on these results, a mesh size of 0.3mm was selected for all 50 simulations. This represented the finest mesh size that could be consistently and reliably produced within the FastX environment, while also displaying a clear trend towards a stable solution for both the inlet pressure and outlet temperature. All 50 geometries simulated using this mesh size successfully reached convergence, confirming the practicality of this mesh size selection for the full DOE process.

3. AI Model Description and Methodology

A machine learning surrogate model was built using the CFD simulation data from the battery cold plate cases. The goal of the model was to reduce computational time while still providing accurate thermal and flow predictions.

The input parameters included channel height (H_c), channel width (W_c), rib thickness (t_r), and coolant mass flow rate $\dot{m}$. The predicted outputs were outlet coolant temperature T_{out} and pressure drop ΔP . The dataset was processed in Python using NumPy and Pandas. Additional parameters such as Reynolds number, hydraulic diameter, and flow velocity were also calculated to improve prediction accuracy.

$$Re = \frac{\rho v D_h}{\mu}$$

XGBoost regression was selected as the machine learning model because it performs well for nonlinear engineering datasets. The dataset was divided into training, validation, and testing groups before training.

Outlier detection methods were also applied before training to remove abnormal data points and improve model stability. Feature engineering was used to generate additional flow-related parameters from the original CFD data.

Model performance was evaluated using MAE, RMSE, and R^2 values. Parity plots and residual plots were also used to compare the AI predictions with the CFD results.

Overall, the surrogate model showed good agreement with the CFD simulations while requiring much lower computational cost.

4. Results and Discussion

This section presents the CFD simulation results and machine learning analysis for the battery cold plate study. First, the effects of the design variables on outlet temperature, pressure drop, pumping power, and thermal resistance are discussed based on the CFD results. Then, the prediction performance of the surrogate model is evaluated and compared using several statistical metrics and residual analyses. These results help assess both the thermal-hydraulic behavior of the cold plate and the accuracy of the AI-based prediction model.

4.1 CFD Trends

For all 50 simulations, an analysis of CFD trends was performed. Four design variables defining the cold-plate configurations included the total coolant mass flow rate, the channel flow width, height, and spacing. From each simulation, three primary outputs were recorded - flow velocity, pressure drop, and outlet coolant temperature. The channel flow cross-section and the required pumping power were calculated. Mass flow rate, as well as the total flow area were used as the input variables for the analysis. Outlet temperature and pumping power were used as the output variables. The trends shown below characterize the thermal and hydraulic performance of the coldplate, and its' sensitivity to these inputs.

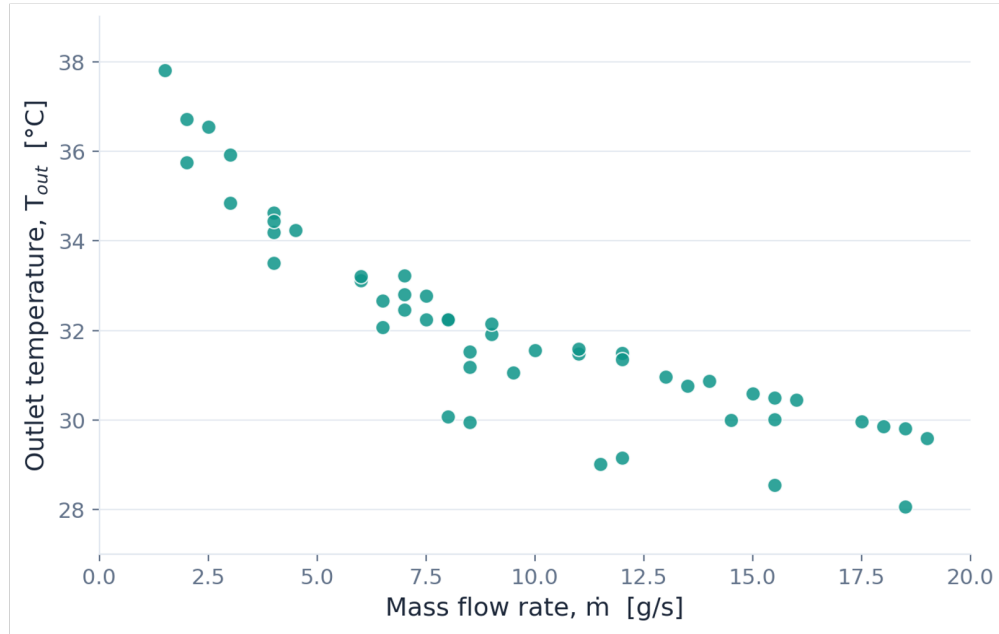


Figure 22: Flowrate dependency (temperature)

Figure 22 plots the outlet coolant temperature against the mass flow rate and shows a clear decrease. A larger mass of fluid passing through the channels carries away the same heat with less temperature increase, so both the coolant and the plate end up being colder. The relationship is clearly nonlinear, showing that after a certain point there is no benefit to the flow rate increase. This shows that increasing the flow rate to the maximum is not an effective strategy due to the limited heat conduction through the cold plate.

Figure 23 shows a plot of pressure drop against the mass flow rate. The behavior here is complementary, the pressure drop increases with increasing flow rate. The first derivative is increasing, showing the growing hydraulic penalty. Higher coolant velocities generate greater frictional loss.

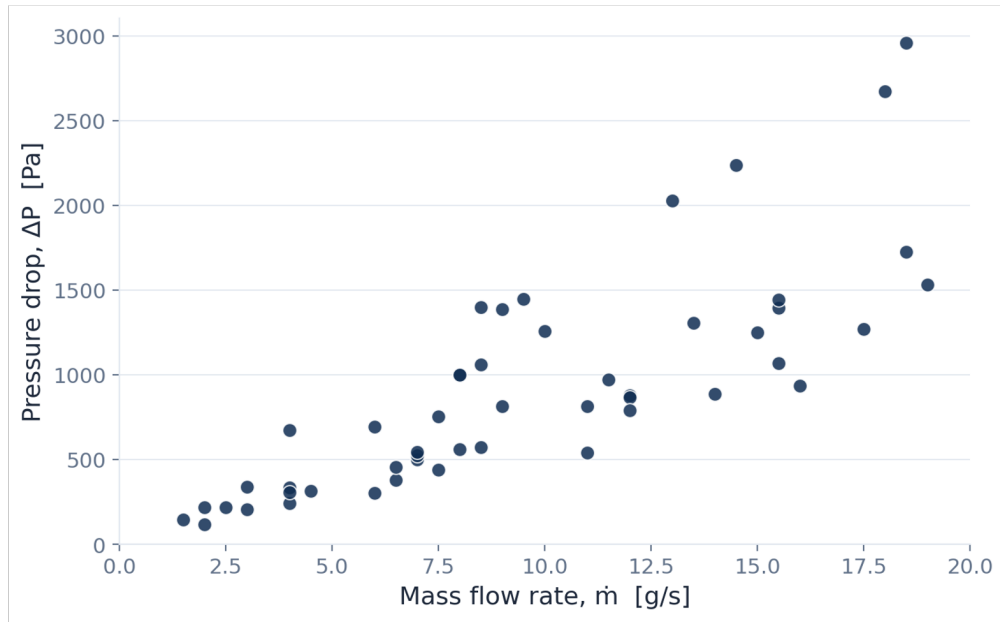


Figure 23: Flowrate dependency (pressure drop)

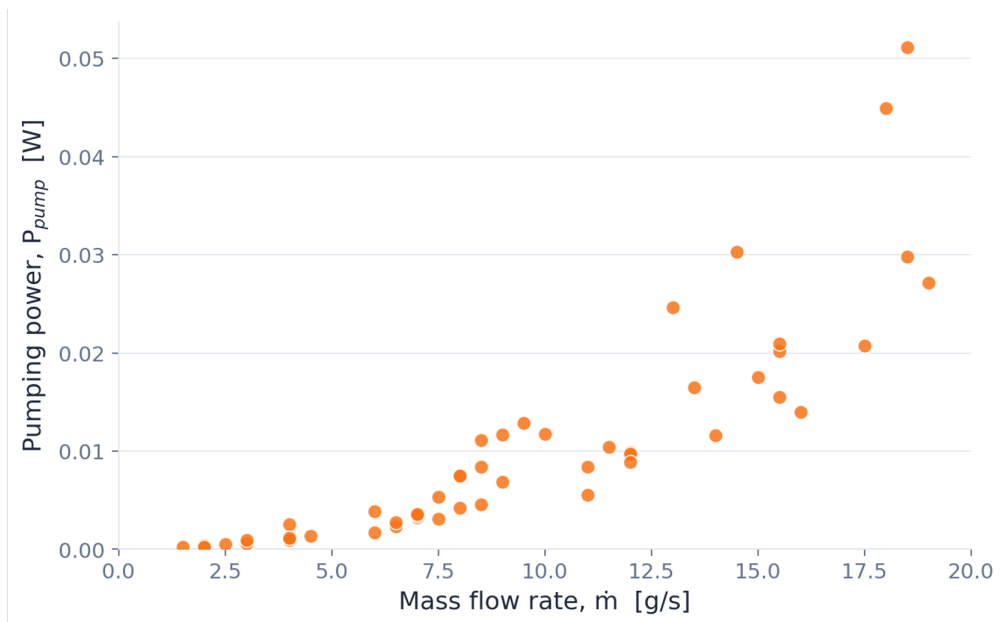


Figure 24: Flowrate dependency (pumping power)

Figure 24 plots the pumping power against the mass flow rate and shows that it increases with flow rate. This trend is very similar to the one shown in Figure 23 due to the direct correlation between pressure drop and pumping power.

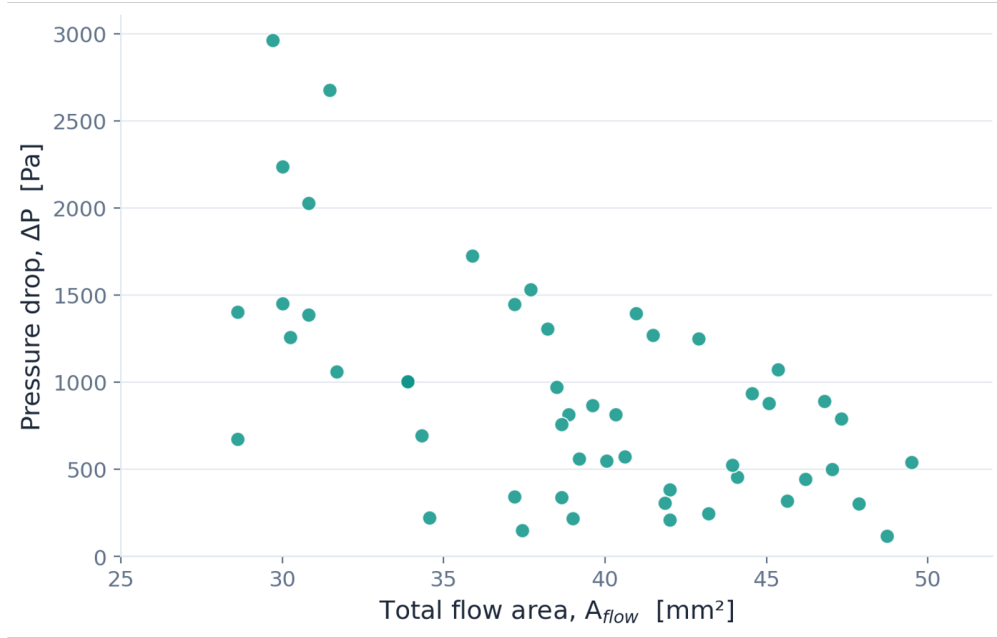


Figure 25: Flowrate area dependency (pressure drop)

Figure 25 shows a plot of pressure drop against the total flow area. The data show that a smaller flow area leads to a larger pressure drop, as constricting the cross-section increases the flow resistance. This plot demonstrates that channel geometry also has a large effect on hydraulic performance.

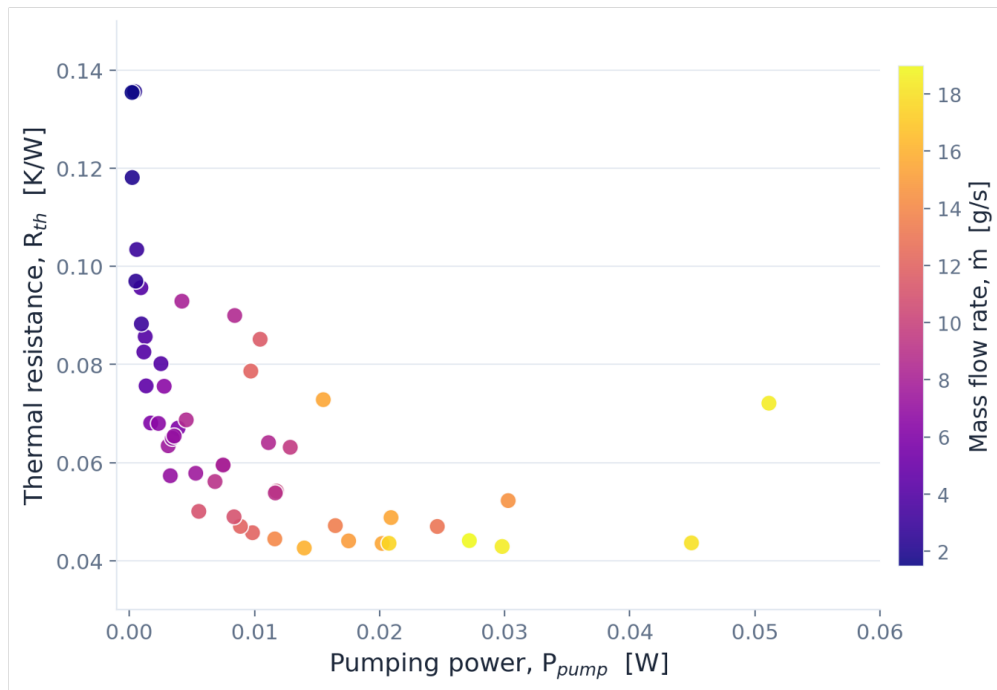


Figure 26: Thermal Resistance

Figure 26 shows a 3-dimensional plot, aiming to combine the thermal and hydraulic sides of the problem. The plot shows thermal resistance vs. pumping power, with each dot colored to show its' mass flow rate. The design goal is to push toward the bottom-left corner of the plot - low thermal resistance and low pumping power. This plot clearly shows why the flow rate needs to be balanced rather than maximized, resulting in acceptable cooling without an unreasonable pumping power.

4.2 ML Comparison

Several statistical metrics were used to evaluate the surrogate-model performance, as summarized in Figure 27.

Metric	Meaning	Interpretation
R^2	Measures how well the prediction follows the CFD trend	Higher R^2 = better agreement and overall prediction performance
MAE (Mean Absolute Error)	Measures the average prediction error magnitude	Lower MAE = prediction is closer to CFD results on average
RMSE (Root Mean Squared Error)	Measures overall prediction error with stronger penalty on large deviations	Lower RMSE = better prediction accuracy and stability
Residual	Prediction – CFD Result	Residual closer to 0 = better agreement with CFD
Std (Standard Deviation)	Measures how spread out the residuals are	Lower Std = more consistent prediction errors

Figure 27: Model evaluation metrics

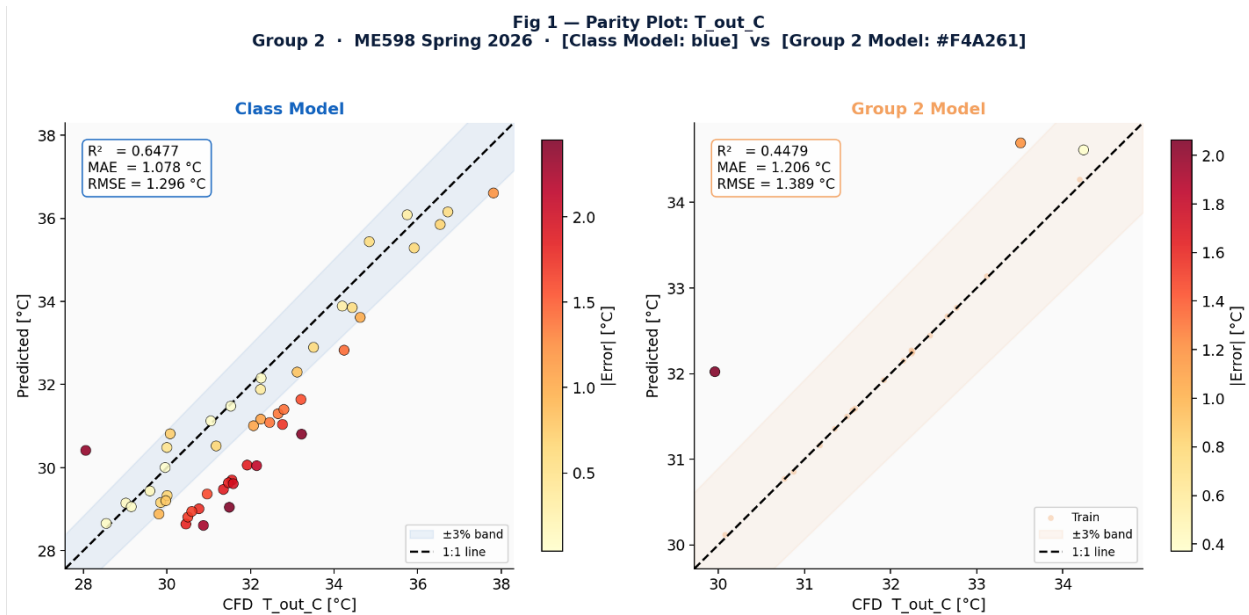


Figure 28: T_out prediction accuracy comparison

Figure 28 compares the outlet-temperature prediction accuracy between the class model and the Group 2 model. The class model achieved a higher R^2 value and slightly lower MAE and RMSE

values, indicating better overall prediction performance. In contrast, the Group 2 model showed more scattered prediction points and larger deviations from the 1:1 reference line. One possible reason is the smaller validation dataset and the limited geometric variation within the Group 2 dataset. Although the prediction accuracy is lower, the Group 2 model still captures the general outlet-temperature trend reasonably well.

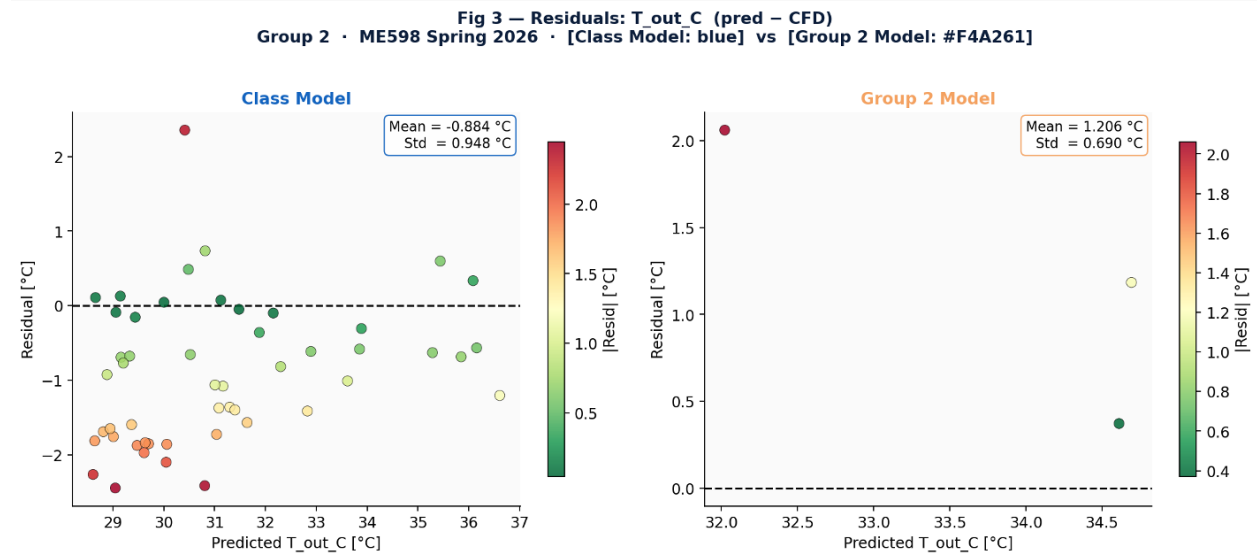


Figure 29: T_out residual comparison

The residual comparison showed that the Group 2 model had a smaller residual spread, indicating more consistent predictions. However, most residuals remained positive, suggesting a systematic overprediction trend for outlet temperature.

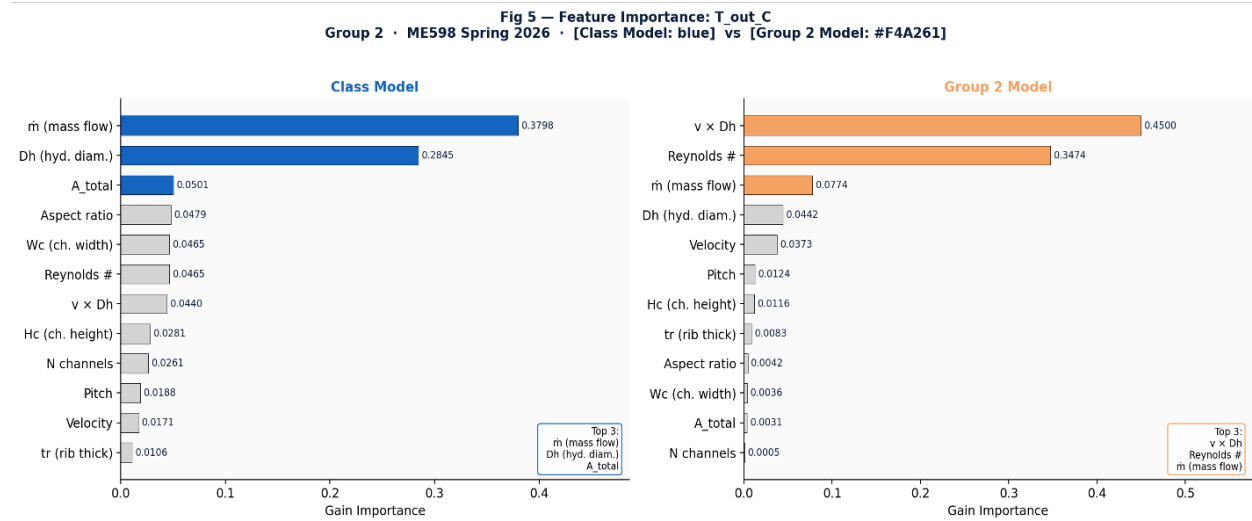


Figure 30: T_out feature importance

Feature-importance analysis also showed that the Group 2 model relied more heavily on combined

flow parameters such as Reynolds number and vD_h , while the class model was more strongly influenced by individual parameters such as mass flow rate and hydraulic diameter.

In addition, outlet temperature is influenced by multiple coupled thermal and flow effects, making it more difficult for the surrogate model to fully capture all local CFD behaviors. Small variations in flow distribution and heat transfer inside the channels may lead to noticeable prediction deviations, especially when the available validation dataset is limited.

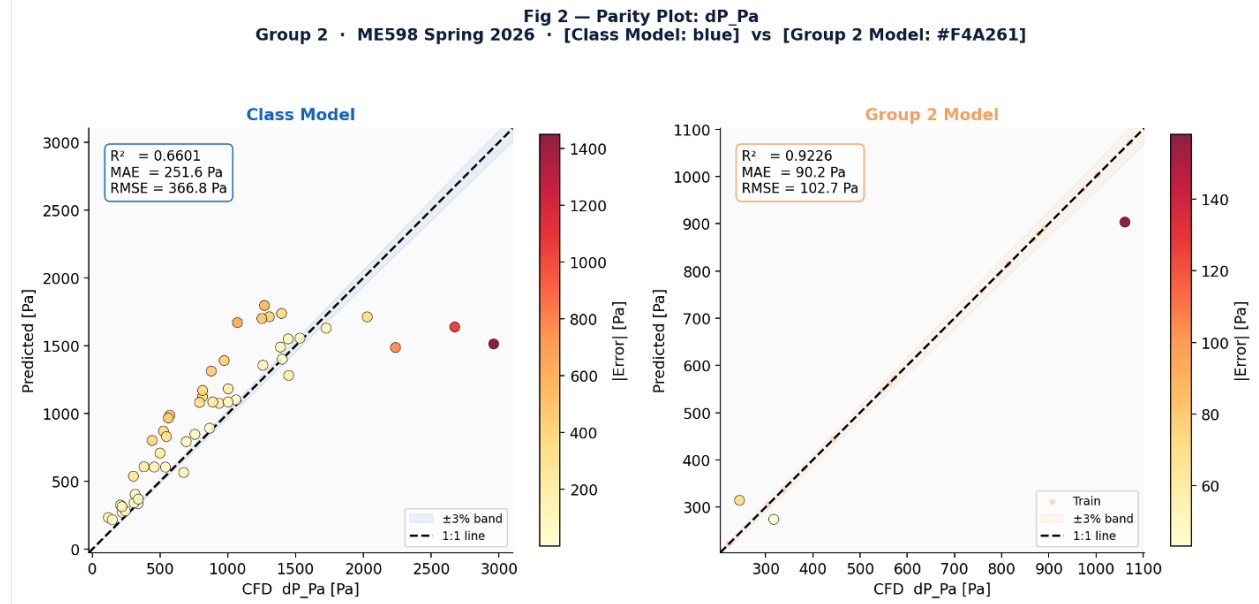


Figure 31: dP Prediction accuracy comparison

Figure 31 compares the pressure-drop prediction accuracy between the class model and the Group 2 model. The Group 2 model achieved a significantly higher R^2 value and lower MAE and RMSE values, indicating much better agreement with the CFD results. Most prediction points were distributed close to the 1:1 reference line, showing that the surrogate model successfully captured the overall pressure-drop trend. Although one larger deviation still appeared at high pressure-drop conditions, the overall prediction performance remained very strong.

The residual comparison in Figure 32 showed that the Group 2 model had a much smaller residual spread than the class model, indicating more stable and consistent pressure-drop predictions. Most residuals were distributed close to zero, showing good agreement with the CFD results. However, a larger negative residual was still observed at high pressure-drop conditions, suggesting reduced prediction accuracy for several extreme cases.

Feature-importance analysis showed that the Group 2 model relied more heavily on flow-related parameters such as Reynolds number, velocity, and channel width. In contrast, the class model was more strongly influenced by channel height. One possible reason is that the Group 2 dataset covered a narrower channel-height range, reducing the sensitivity of the model to geometric variations. As a result, flow-related parameters became more dominant for representing the pressure-drop behavior within the local dataset.

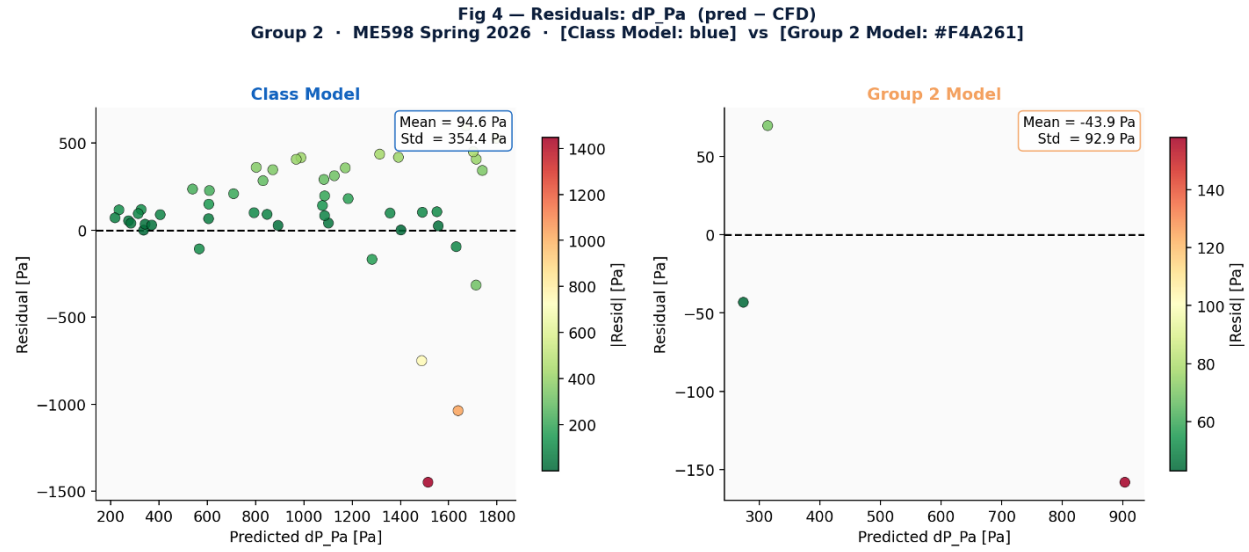


Figure 32: dP residual comparison

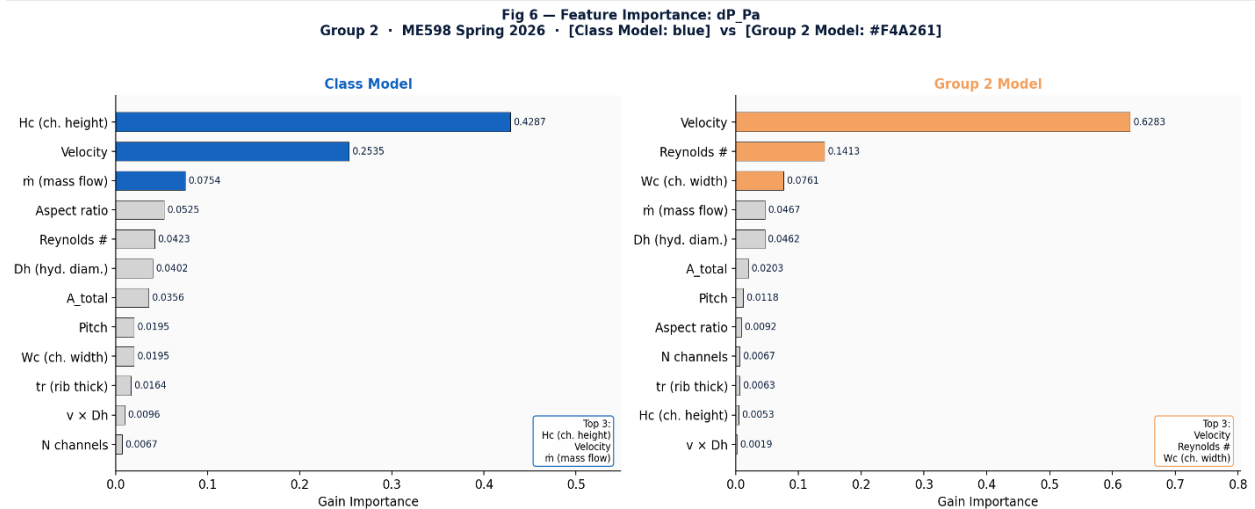


Figure 33: dP feature class importance

In addition, pressure drop follows a clearer flow-resistance relationship than outlet temperature, making the overall trend easier for the surrogate model to learn. Since pressure drop is highly dependent on velocity and flow resistance inside the channels, the AI model was able to capture the CFD trend more accurately.

Figure 34 compares the overall R^2 performance of the two surrogate models. For outlet-temperature prediction, the class model achieved a higher R^2 value, indicating better overall prediction accuracy. In contrast, the Group 2 model performed significantly better for pressure-drop prediction, with an R^2 value above 0.90. This result suggests that the Group 2 surrogate model is more reliable for predicting pressure-drop behavior than outlet temperature.

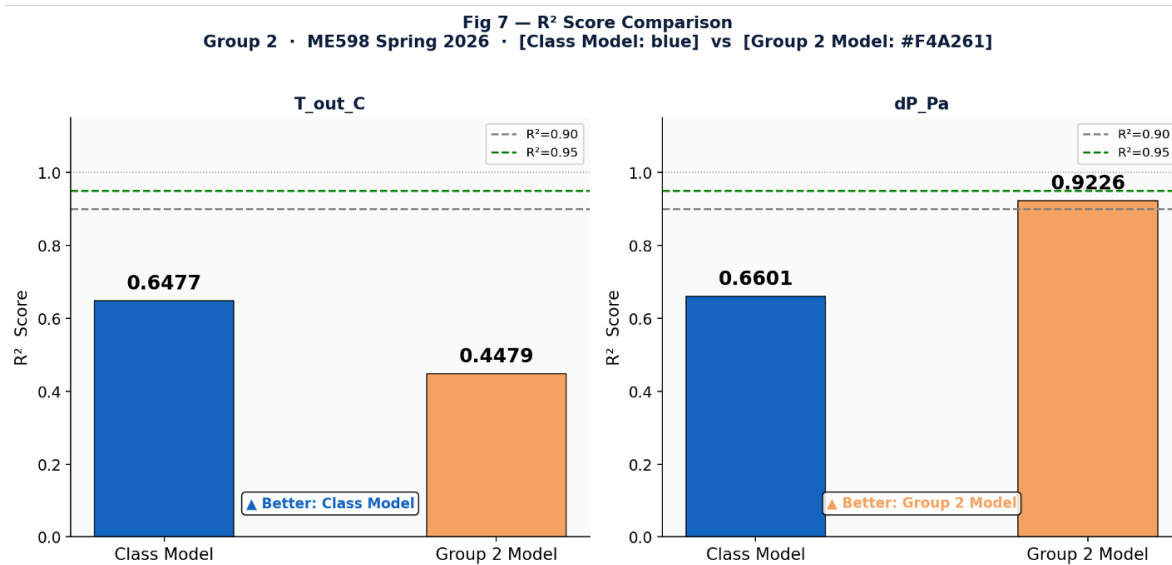


Figure 34: Model performance R² analysis

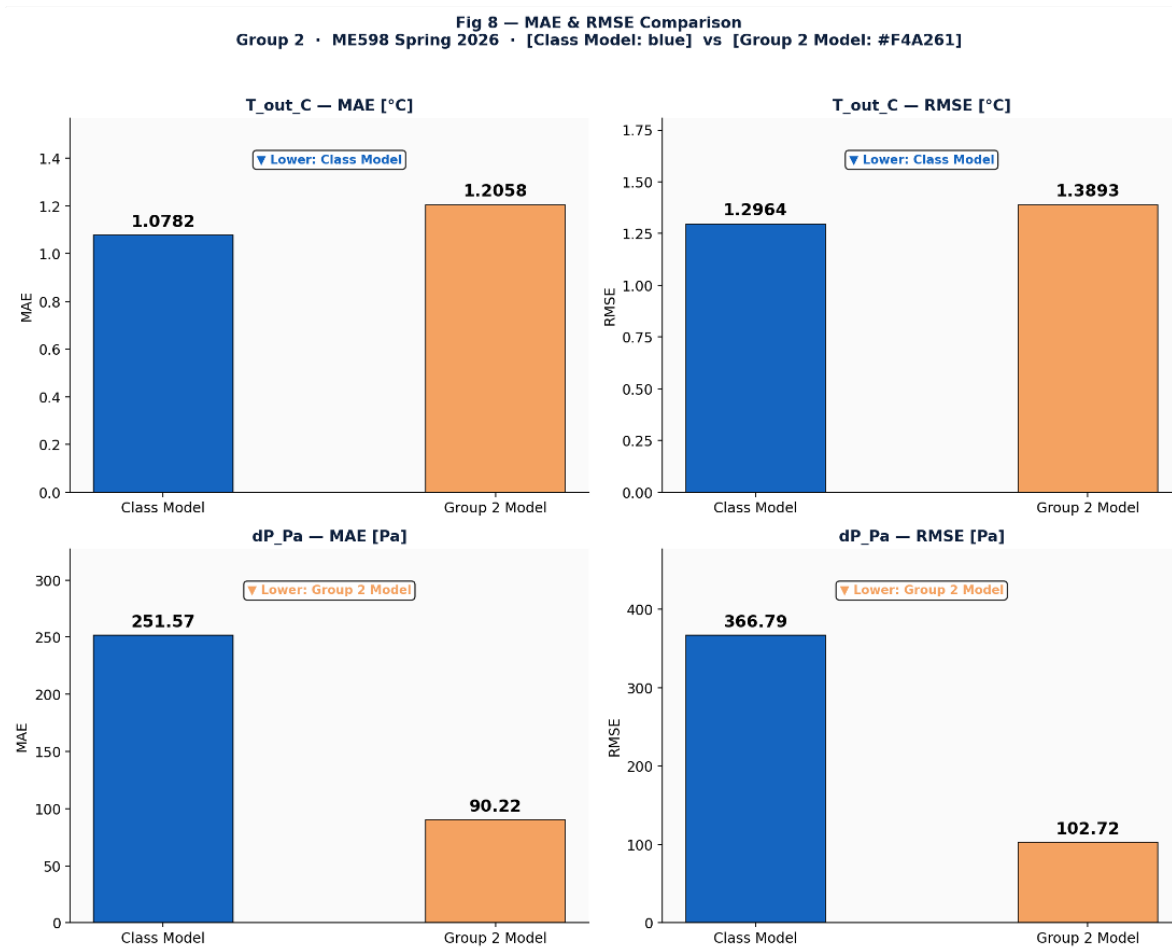


Figure 35: Model Performance MAE & RMSE analysis

Figure 35 compares the MAE and RMSE values for both outlet-temperature and pressure-drop predictions. The differences between the two models are relatively small for outlet temperature. However, the Group 2 model achieved much lower prediction errors for pressure drop, indicating stronger agreement with the CFD results. Overall, the surrogate model showed better performance for pressure-drop prediction than for outlet-temperature prediction.

5. Conclusion

5.1 Summary of Findings

This study successfully demonstrated an integrated CFD–machine learning framework for the design and optimization of an EV battery liquid cold plate. The workflow enabled systematic exploration of the thermal–hydraulic design space while significantly reducing computational cost through surrogate modeling.

From a fluid–thermal perspective, the simulations confirmed that the operating regime remains fully laminar across all design conditions ($Re < 500$), ensuring predictable flow behavior and stable numerical convergence. This simplifies physical interpretation, as pressure losses are dominated by viscous effects and heat transfer is governed primarily by conduction-convection coupling. Clear design trends emerged from the parametric study. Increasing mass flow rate improves heat removal and reduces outlet temperature, but with diminishing returns due to reduced thermal gradients at higher flow rates. At the same time, pressure drop and pumping power increase strongly with flow rate, creating a fundamental tradeoff between thermal performance and hydraulic cost. Similarly, reducing flow area (through smaller channel dimensions) increases flow resistance, further amplifying pressure losses. These competing effects highlight the inherently multi-objective nature of cold plate design. An optimal configuration was identified that balances these tradeoffs, achieving low thermal resistance (~ 0.05 K/W) while maintaining low pumping power (~ 0.0055 W). This demonstrates that careful tuning of geometric parameters (channel height, width, and rib thickness) alongside flow rate can yield high-performance designs without excessive parasitic losses.

From a system-level perspective, scaling analysis showed that while pumping power is relatively small compared to other vehicle loads (e.g., HVAC), it can vary by an order of magnitude depending on design choices. This indicates that although cold plate pumping energy may have a limited direct impact on vehicle range, poor hydraulic design can still lead to unnecessary inefficiencies, particularly under high-load conditions such as fast charging.

The surrogate modeling results demonstrated that machine learning models can effectively approximate CFD predictions, with performance varying by output variable. Pressure drop predictions showed strong agreement with CFD results ($R^2 \approx 0.92$), low error, and tight residual distributions, indicating high model reliability. In contrast, outlet temperature predictions exhibited greater scatter and lower accuracy ($R^2 \approx 0.45\text{--}0.65$), reflecting higher sensitivity to input parameters and potentially insufficient coverage of the design space. Feature importance analysis further revealed that flow-related parameters (velocity, Reynolds number, and combined flow metrics) play a dominant role in predicting both thermal and hydraulic behavior, especially within narrower design ranges. This suggests that the surrogate models capture the governing physics effectively, but also highlights the importance of dataset diversity to decouple correlated variables.

Finally, manufacturability considerations showed that while most channel geometries are feasible using conventional methods (e.g., extrusion and brazing), rib aspect ratios approach practical limits. This underscores the importance of integrating design-for-manufacturing constraints early in the optimization process. Overall, the study validates that AI-assisted CFD workflows can significantly accelerate thermal system design, enabling rapid exploration of complex, multi-objective problems while maintaining acceptable accuracy.

5.2 Future Work

While the current study provides a strong foundation, several opportunities exist to enhance both the physical modeling fidelity and the machine learning framework.

A key extension is to improve surrogate model accuracy for thermal predictions, particularly outlet temperature. This can be achieved by increasing the size and diversity of the training dataset, especially by including more validation points and expanding the parameter space. Incorporating additional physics-informed features (e.g., Nusselt number correlations or thermal boundary layer metrics) could further improve predictive capability.

Another important direction is to move beyond steady-state assumptions and develop transient models that capture dynamic battery operation, such as fast charging cycles, variable load conditions, and thermal cycling. This would significantly improve the applicability of the model to real-world EV scenarios.

From a CFD standpoint, future work could incorporate more complex physics, including turbulence modeling for higher flow regimes, non-uniform heat generation within battery cells, and temperature-dependent material properties. Additionally, modeling full cooling loop interactions, including manifolds, connectors, and multiple cold plates, would provide a more realistic system-level perspective.

The integration of multi-objective optimization algorithms (e.g., genetic algorithms or Bayesian optimization) with the surrogate models represents another promising direction. This would enable automated identification of Pareto-optimal designs balancing thermal resistance, pressure drop, manufacturability, and cost.

Further, incorporating manufacturing constraints directly into the optimization loop would improve design feasibility. This includes constraints on rib thickness, aspect ratios, and allowable tolerances based on specific fabrication processes such as extrusion or brazing.

On the machine learning side, exploring advanced modeling techniques such as neural networks, ensemble methods, or physics-informed neural networks (PINNs) could improve generalization and robustness, especially for nonlinear thermal behavior. Active learning strategies could also be employed to iteratively select new CFD simulation points that maximize model improvement.

Finally, a critical next step is experimental validation. Building and testing a prototype cold plate under controlled conditions would allow direct comparison with CFD and surrogate predictions, providing confidence in the model's real-world applicability and identifying gaps in the current assumptions.

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